C2RCC algorithm recalibration for the assessment of Chlorophyll-a concentration in shallow coastal lagoons

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ABSTRACT

Chlorophyll-a (Chl-a) is one of the main biological parameters measured to assess the ecological status in aquatic systems. Remote sensing has been an effective tool for the systematic monitoring of water quality, recording surface Chl-a levels and gaining knowledge on eutrophication dynamics. In Sentinel 2 image analysis, the SNAP software and the C2RCC processor are being widely used for Chl-a assessment. However, in the highly productive, turbid and shallow coastal lagoons this algorithm fails to produce valid results and recalibration is needed. In-situ Chl-a concentration data sampled from the Nestos coastal lagoons and the respective reflectance values for bands 4-7 were imported in a Takagi-Sugeno neuro-fuzzy model to improve Chl-a assessment. Based on these results, the spatio-temporal variability of Chl-a concentration for Eratino lagoon (Nestos lagoon complex) was assessed to determine the current ecological status from 2015 to 2021.

Keywords: chlorophyll-a, eutrophication, remote sensing, Takagi-Sugeno neuro-fuzzy model, Sentinel 2.

1. Introduction

Consistent and cost-efficient monitoring in coastal ecosystems is essential, but in situ monitoring (e.g., water sample collection and laboratory analysis) is a time and money consuming method for estimating the quality of water on a regular basis. Satellite remote sensing is a feasible and cost-effective method to monitor water bodies when water quality over large regions has to be monitored with regular frequency. There is also the possibility to estimate water quality in non-accessible water bodies. However, passive satellite remote sensing depends highly on weather, like cloudiness, air mass changes, and sunlight conditions, which directly affect the quality and quantity of data. In parallel, the current algorithms for SPM and Chl-a concentration evaluation based on spectral reflectances, like the C2RCC for Sentinel 2, have been validated in the open sea environment of Case 1 (in which their inherent optical properties are dominated by phytoplankton, e.g., most open ocean waters) and Case 2 waters (containing coloured dissolved organic matter (CDOM) and inorganic mineral particles in addition to phytoplankton (Ansper and Alikas 2019, Soomets et al., 2020). However, the interference of (a) the shallow sea bottom reflection, (b) the sun glint and (c) the presence of non-algal particles on the optical signal (spectral reflectances) measured by satellites has not been adequately evaluated. Yu et al. (2020) developed a semi-empirical algorithm to correct Chl-a concentration observed by SPOT6 at the very shallow parts of Sanya Bay, China. In the present work, we attempt to recalibrate the C2RCC processor using in-situ Chl-a concentration data and the respective spectral reflectance values for the appropriate training of a Takagi-Sugeno neuro-fuzzy algorithm to correct the satellite-derived Chl-a values.

2. Methodology

A series of Sentinel 2 historic satellite images were retrieved for the study area (Eratino lagoon, Nestos delta, Northern Greece) to cover the period from early 2015 to late 2021. Eratino Lagoon belongs to the Nestos complex, part of the East Macedonia and Thrace National Park. The area is protected by the Convention on Wetlands of International Importance (Ramsar Convention). It covers an area of 2.9 km^2, about 5.9 km long, 0.7 km wide and has a perimeter of 43 km. The mean depth of the lagoon is 0.8 m and the maximum depth is 3.4 m.

All Sentinel 2 images were retrieved from the L1C product containing reflectances with the same atmospheric correction. To derive the Chl-a concentration, retrieved images were processed by Sentinel Toolbox Development Platform (SNAP) v.8.0 software, using the C2RCC processor (Brockman et al., 2020). The C2RCC processor is developed for optically complex Case 2 waters. It is based on neural network technology and has been trained under extreme ranges of scattering and inherent absorption properties (IOPs). The C2RCC results of Chl-a concentration was derived, after the provision of additional background data in parameters such as water surface salinity, elevation, ozone, water surface temperature and air pressure. Salinity and temperature values were used from previous in situ data and values of the surface ozone layer and air pressure data were retrieved from ERA5 model with 30 km spatial resolution.

A database of 122 in-situ Chl-a concentration values, collected from the shallow parts of Nestos lagoons complex were associated with the respective reflectance values from bands 4 to 7 of Sentinel 2. 60% of data from the database were used for training the Takagi-Sugeno neuro-fuzzy model, 20% for model validation and 20% for model testing. The model was implemented in Matlab 2022 utilizing the standard Adaptive Neural Fuzzy Inference System (ANFIS) algorithm. Initially, the fuzzification of input values (4 antecedents: reflectances at each band; 1 output: log(Chl-a)) through the general bell-shaped membership function and the grid partition method, led to the definition of membership values in the three fuzzy set ("low", "medium", "high"). Then, a series of multiple-input–single output fuzzy rules were applied of the form: *if band1 is "LOW" AND band2 is "MEDIUM" AND band3 is "LOW" and band4 is "HIGH" then log(Chl-a) = f(band1, band2, band3, band4)*. Finally, the evaluation and weighting of the basic functions and the final evaluation of the output Chl-a value were followed in the defuzzification step. Model testing error analysis exhibited the following metrics: Mean Squared Error = 0.000018; Root Mean Squared Error = 0.00418; r^2 = 0.996. Using this neuro-fuzzy model all Sentinel 2 Chl-a data were corrected for Eratino lagoon.

3. Results

Figure 1 illustrates the Chl-a concentration distribution for 19 July 2021, as derived by applying the C2RCC processor (left panel) and the neuro-fuzzy model (right panel). C2RCC processor estimated high Chl-a values ranging from 15-30 μg/l. The highest values were found at the northern and eastern parts of the lagoon. The lowest values (close to zero) were estimated at the freshwater inflow stream, at the northern part of the basin. As this instream supplies nutrients from agricultural runoff, Chl-a values were expected to be high. The Chl-a values adapted after the neuro-fuzzy model application were significantly lower (up to 3 μg/l). The neuro-fuzzy model seems to capture the expected higher values at the instream flow (reaching 30 μg/l).

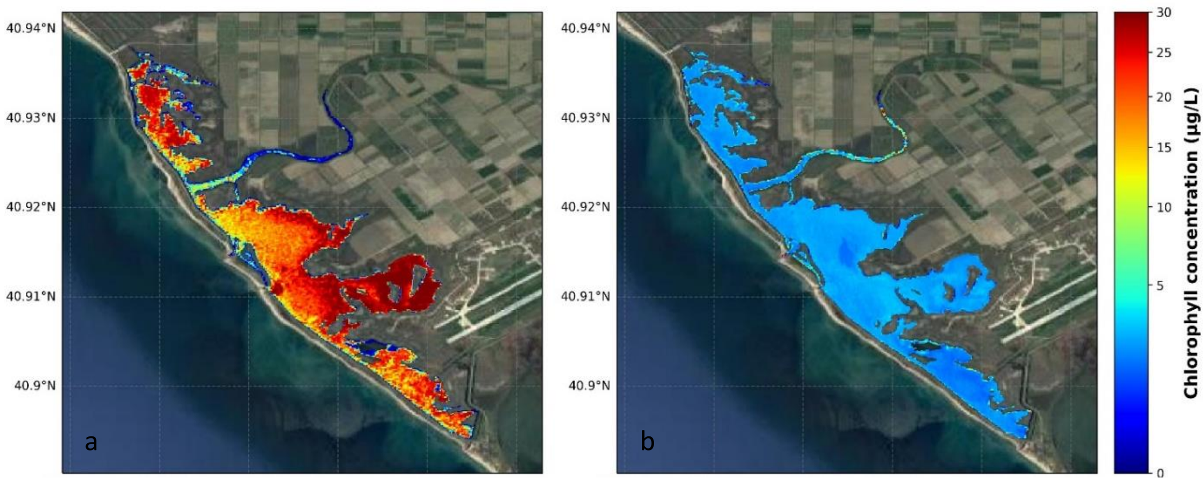

Fig. 1. Spatial evolution of Chl-a in Eratino lagoon (19/07/2020) derived using a) the C2RCC processor and b) the neuro-fuzzy model.

4. Conclusions

In this work, a neuro-fuzzy model was developed to correct the observed by the Sentinel 2 satellite Chl-a values. The algorithm was implemented at Eratino Lagoon (Northern Greece), giving satisfactory results.

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